Peiyuan Chen

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Research Profile

Detector instrumentation researcher working on PMT acquisition, characterization, waveform correction, and position reconstruction for liquid-xenon rare-event experiments.

Education

Jilin University — Changchun, China
Bachelor of Science, Physics — Sept 2020 – June 2026

- Rank in major: 8/29.
- B.S. thesis: Development and Testing of R12699 and R11410 PMT Arrays: A Comparative Study of Position-Reconstruction Algorithms. Outstanding Undergraduate Thesis Award, Jilin University.

Georgia Institute of Technology — Atlanta, USA
Visiting Undergraduate Student, Physics — Jan 2024 – May 2024

Research Experience

PandaX Collaboration, TDLI, Shanghai Jiao Tong University, Research Assistant — Shanghai, China
June 2024 – present

- Led development and maintenance of a 192-channel universal PMT signal acquisition platform, integrating high voltage, signal readout and separation, 500 MS/s waveform digitization, over 1,000 MB/s readout bandwidth, and storage; operated stably for more than 500 days.
- Established waveform-processing workflows and a performance database for dark count, gain, and afterpulse evaluation; characterized more than 160 R11410 and 500 R12699 PMTs at ambient and low temperatures.
- Evaluated rigid-flex PMT bases and informed trace-spacing and ground-isolation improvements. Noise and attenuation fell to about 1/5 and 1/3 of their original levels; average crosstalk fell from 15–20% to about 1%, and maximum crosstalk from 28% to below 3%.
- Measured nonlinear PMT response over approximately 1–100,000 PE per channel and pulse separations of 5–100 µs; co-first author of the R12699 base study in IEEE Transactions on Nuclear Science.
- Developed a coupled base-circuit and custom 2.5D electron-transport diagnostic model for saturation and suppression.
- Applied waveform corrections to detector data; using the same event selection before and after correction, reduced the difference between multi-site and single-site reconstructed energy peak positions by 85%.
- Led hardware, DAQ, calibration, Geant4 optical simulation, and reconstruction analysis for a 55-channel R11410 array; achieved approximately 0.66 mm overall spatial resolution in array performance tests.

Georgia Institute of Technology, Research Intern (Trinity PeV Neutrino Observatory) — Atlanta, USA
Mar 2024 – May 2024

- Designed and fabricated detector connection components for the Trinity PeV Neutrino Observatory.

PandaX Collaboration, Research Assistant — Shanghai, China
Jan 2022 – Mar 2022

- Designed a calibration device for the PandaX-III high-pressure gaseous-xenon TPC with Micromegas readout; reduced thickness while maintaining mechanical strength and leak-tightness, achieving sub-millimeter position precision.

Publications

Performance Study and Circuit Simulation for R12699-406-M4 Photomultiplier Tube Base June 2026
IEEE Transactions on Nuclear Science 73(6), 2511-2519. Published; co-first author with Houqi Huang.

Measurement of Solar pp Neutrino Flux with the New PandaX-4T Data July 2026
arXiv:2607.02405. PandaX Collaboration paper; author list is ordered alphabetically.

Bi-Interlayer Strategy for Modulating NiCoP-Based Heterostructure toward High-Performance Aqueous Energy Storage Devices July 2024
Advanced Materials. Co-author; contributed to experiments and simulations.

Manuscripts and Thesis

Coupled PMT Electron-Transport and Base-Circuit Modeling for Saturation and Suppression Correction in Liquid-Xenon Detector Data
Working title; manuscript in preparation.

Development and Testing of R12699 and R11410 PMT Arrays: A Comparative Study of Position-Reconstruction Algorithms 2026
B.S. thesis, Jilin University; Outstanding Undergraduate Thesis Award.

Projects

192-Channel PMT Acquisition and Characterization
500 MS/s digitization; over 1,000 MB/s readout bandwidth; more than 160 R11410 and 500 R12699 PMTs characterized.

PMT Waveform Correction
Measured response and diagnostic modeling applied to detector data; 85% reduction in MS–SS reconstructed energy peak separation under the same event selection.

55-Channel R11410 Array and Position Reconstruction
Hardware, DAQ, calibration, optical simulation, and reconstruction; approximately 0.66 mm spatial resolution in array performance tests.

Rigid-Flex PMT Base Evaluation
Thermal, connectivity, noise, attenuation, and crosstalk tests informing layout improvements.

Awards

First Prize, Annual Review of Nuclear and Particle Science Video Contest 2026
Annual Reviews.

National First Prize, China Undergraduate Physics Experiment Competition (CUPEC) 2023
Ranked 1st in comprehensive experimental problem D.

High-Level Academic Competition Achievement Scholarship, Jilin University 2025

Student of the Year (Group Member), Shanghai Jiao Tong University 2023

13th Place, China Undergraduate Physics Tournament 2021

Grand Prize, Northeast Division Undergraduate Physics Tournament 2021

Patents

Small Wind Tunnel Device Used in Laboratory
Chinese patent CN215414302U. Contributed to design, fabrication, and testing.

Method of Measuring Wind Speed Using a Constant Flow Thermal Anemometer
Chinese patent CN113466487B. Contributed to design, fabrication, and testing.

Skills

Programming & Data: Python, C++, MATLAB, ROOT, Geant4, Git, LaTeX

Experimental: PMT characterization, detector readout, DAQ systems, circuit design, vacuum systems

Computation: Monte Carlo simulation, finite element analysis, machine learning for reconstruction

Languages

Chinese: Native

English: Fluent

Leadership and Service

CUPT Community, Jilin University, Founder & Chairman — Changchun, China — Feb 2021 – June 2023

- Founded and led an approximately 30-member CUPT community with a five-member executive committee.
- Delivered five training sessions for audiences of 20–100 on instrument use, experimental design, uncertainty analysis, and data analysis; trainees from that cohort won a national first prize.

JLUPT, Chairman & Referee — Changchun, China — May 2022 – May 2022

- Organized and judged university physics tournament